Luis Valenzuela Cazares

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github.com/luisval · https://gitlab.com/luisval · Portfolio

Education

Iowa State University — **Aug 2020 – May 2023**
Master of Science in Nuclear Physics — *Ames, Iowa*

Universidad de Sonora — **Aug 2017 – Aug 2020**
Master of Science in Physics — *Hermosillo, Mexico*

Universidad de Sonora — **Aug 2011 – May 2016**
Bachelor of Science in Physics — *Hermosillo, Mexico*

Research Profile

Physicist with research experience in experimental nuclear physics through collaborations at Brookhaven National Laboratory and Fermilab. Experienced in detector development, experimental data analysis, and computational simulations, with research interests in atomic and nuclear physics, precision spectroscopy, and detector *R&D*.

Research Experience

Graduate Research Assistant — **Feb 2023 – April 2023**
Iowa State University and Brookhaven National Laboratory — *Upton, NY*

- Developed mathematical and computational methods to optimize the firing pattern of the TPC calibration lasers in the sPHENIX experiment using the detector simulation framework.
- Performed detector-performance studies for the sPHENIX TPC laser calibration system to improve charged-particle tracking and momentum reconstruction.
- Contributed to the sPHENIX simulation framework through software development, code integration, and collaborative version control within the experiment's Git repository.

Graduate Research Assistant — **May 2021 – May 2023**
Iowa State University — *Ames, IA*

- Conducted detector-performance and particle-flow analyses for the sPHENIX experiment by combining electromagnetic and hadronic calorimeter measurements to evaluate reconstructed observables in high-energy collisions.
- Participated in the assembly, testing, and calibration of the sPHENIX hadronic calorimeter and conducted cosmic-ray calibration studies contributing to detector commissioning and performance validation.
- Developed and evaluated jet-reconstruction algorithms for the ePIC experiment by comparing Monte Carlo truth information with detector-level observables for future Electron–Ion Collider feasibility studies.
- Developed particle-identification and pion background-rejection techniques for the ePIC experiment through calorimeter-response analysis, achieving approximately 90% pion-background suppression for lepton identification in deep-inelastic scattering studies.

Research Student — **Aug 2020 – June 2021**
Iowa State University and Fermilab — *Ames, IA*

- Implemented a Keras-based machine-learning algorithm for neutron-background discrimination in the ANNIE experiment, improving signal identification and detector-response characterization.

Research Student — **May 2016 – Aug 2020**
Universidad de Sonora and Joint Institute for Nuclear Research (JINR) — *Hermosillo, Mexico*

- Conducted detector *R&D* and physics analysis for the MPD experiment by studying charged-particle observables and detector response in simulated heavy-ion collisions using the MPDROOT framework.
- Mentored undergraduate and graduate students through coding workshops, simulation tutorials, and seminars on computational methods in nuclear physics.
- Developed numerical and computational methods for heavy-ion collision simulations and detector-response studies used in event characterization and centrality analysis.
- Regular presentations in group meetings and international conferences for diverse audiences.
- Participated in collaborative software development and maintenance of the MPD-NICA simulation framework repository.

Research Visitor — Feb 2019 – May 2019
Joint Institute for Nuclear Research — *Dubna, Russia*
- Conducted feasibility studies for centrality determination in the MPD experiment through multiplicity and detector-response analyses in simulated heavy-ion collisions.

JINR Summer Student Program — July 2017 – Sep 2017
Joint Institute for Nuclear Research — *Dubna, Russia*
- Conducted heavy-ion collision simulations with the UrQMD Monte Carlo model to study centrality observables and detector response for a Beam-Beam Counter detector within the MPDROOT framework.

Research Visit — Jan 2017 – March 2017
Benemérita Universidad Autónoma de Puebla (BUAP) — *Puebla, Mexico*
- Conducted simulations of heavy-ion collisions using the HIJING Monte Carlo program and received training in ROOT for data analysis and visualization.

Research Summer Student Program at BUAP — Aug 2014
Benemérita Universidad Autónoma de Puebla — *Puebla, Mexico*
- Contributed to the development of the Cosmic Piano, a scintillator-based muon detector that transforms cosmic-ray signals into interactive light and sound displays for science outreach, through initiatives associated with the ALICE Collaboration.

Teaching Experience

Math Teacher — Aug 2023 – Present
Toltecalli High School, CPLC Community Schools — *Tucson, AZ*
- Teach Mathematics, Physics, and Programming courses to students from historically marginalized and underserved populations.
- Developed engaging and differentiated curricula aligned with district and state standards for STEM classes.
- Increased student proficiency on district math benchmarks from 0% to over 30% grade-level performance or beyond.
- Conducted credit recovery classes and tutoring sessions to support at-risk students, improving academic outcomes.

Graduate Teaching Assistant — Aug 2020 – May 2021
Iowa State University — *Ames, IA*
- Taught physics classes and laboratories to students of different academic disciplines.
- Led tutoring and discussion sessions among students, promoting a collaborative environment.
- Developed teaching plans and graded homework, quizzes, and laboratory reports.

Publications and Collaboration Papers

- L Valenzuela Cazares (2023). *Firing Pattern for the TPC Calibration Lasers in the sPHENIX Experiment.* DOI: 10.31274/cc-20240624-740
- VD Burkert et al. (2023). *Precision Studies of QCD in the Low Energy Domain of the EIC.* ArXiv:2211.15746v3
- V Abgaryan et al. (2022). *Status and Initial Physics Performance Studies of the MPD Experiment at NICA.* ArXiv:2202.08970
- RA Kado et al. (2020). *Conceptual Design of the miniBeBe Detector Proposed for NICA-MPD.* ArXiv:2007.11790
- A Ayala et al. (2020). *Core meets corona: a two-component source to explain Λ and Λ^- global polarization in semi-central heavy-ion collisions.* ArXiv:2003.13757
- M Alvarado et al. (2019). *Time resolution studies for scintillating plastics coupled to silicon photomultipliers.* ArXiv:1901.04964

Talks

- Laser alignment optimization for the TPC detector at the 2nd Commissioning Workfest for the sPHENIX experiment (2023).
- Physics simulations with the MPDROOT framework at the 2nd Computing and Analysis Workshop of MexNICA Collaboration (2020).
- Centrality determination of heavy-ion collisions in the MPD-NICA experiment at the Heavy Second Colima Winter School on High Energy Physics at the Universidad de Colima (2018).
- Physics simulations in the MPD-NICA experiment at the seminar of the Particles and Cosmology group of the Universidad de Sonora (2018).
- Monte Carlo workshop at the seminar of the Particles and Cosmology group of the Universidad de Sonora (2018).

Awards and recognition

- Nominated as one of the lead teachers at Chicanos Por La Causa Community Schools (2024).
- Awarded a research visit at the Joint Institute for Nuclear Research as part of the Summer Student Program competition (2017).
- Awarded as the second-best group presentation in the CERN-Latin-American School of High-Energy Physics (CLASHEP) (2017).
- Finalist in the National Nuclear Physics Summer School (NNPSS) contest (2016).
- Finalist in the theoretical-research summer Mexican contest on High Energy Physics (2016).
- Third place, XIX and XX Mexican Chemistry Olympiads (2010, 2011).

Organizational Involvement and Outreach Activities

- Led STEM outreach initiatives at Toltecalli High School, expanding student and family engagement in hands-on science activities within CPLC Community Schools (2024–2025).
- Contributed to software development and detector-performance studies for the sPHENIX TPC laser calibration system during the sPHENIX Commissioning Workfest (2023).
- Directed and led training in simulation and software for the Computing and Analysis Workshops of the MexNICA Collaboration (2020).
- Organized seminar activities and academic events for the Particles and Cosmology group at the Universidad de Sonora (2017–2020).
- Organizer of the Physics Film Club at Universidad de Sonora (2016–2017).
- Responsible for coordinating participant reception and transportation logistics for international CLASHEP attendees (2017).
- Outreach presentation on a cosmic-ray detector (Cosmic Piano) and the muon paradox at *Noche de las estrellas* (2014).

Skills

Programming Languages: C++, Python, Fortran, Shell Scripting, SQL
HEP Tools: ROOT, Geant4, MPDROOT, sPHENIX Core Software, EIC Software, AliRoot, HIJING, UrQMD
Computational Physics: Collider data analysis, detector-response studies, reconstruction algorithms, Monte Carlo simulations, statistical modeling, machine learning, high-performance computing, data visualization
Technical Tools: Linux, Git/GitHub, Jupyter, MATLAB, Microsoft Office Suite

Selected Master's Coursework

- **Quantum Field Theory I & II**
- **Particle and Nuclear Physics**
- **Electricity and Magnetism I & II**
- **Mathematical Methods for the Physical Sciences**
- **Quantum Physics I & II**
- **Statistical Mechanics**
- **Numerical Methods**